Чебышевская сетка:

$$x_k = \frac{a+b}{2} + \frac{b-a}{2}\cos\left(\frac{2k+1}{2n}\pi\right) \Rightarrow x_k = \cos\left(\frac{2k+1}{2n}\pi\right)$$

$x_0 = -\frac{\sqrt{3}}{2}$: $f\left(-\frac{\sqrt{3}}{2}\right) = \ln\left(\frac{7}{4}\right) - \frac{3\sqrt{3}}{8}$

$x_1 = 0$: $f(0) = 0$

$x_2 = \frac{\sqrt{3}}{2}$: $f\left(\frac{\sqrt{3}}{2}\right) = \ln\left(\frac{7}{4}\right) + \frac{3\sqrt{3}}{8}$

$$f(x_0, x_1) = \frac{0 - \left(\ln\frac{7}{4} - \frac{3\sqrt{3}}{8}\right)}{0 - \left(-\frac{\sqrt{3}}{2}\right)} = \frac{-\ln\frac{7}{4} + \frac{3\sqrt{3}}{8}}{\sqrt{3}/2}$$

$$f(x_1, x_2) = \frac{\ln\frac{7}{4} + \frac{3\sqrt{3}}{8}}{\sqrt{3}/2}$$

$$f(x_0, x_1, x_2) = \frac{\frac{\ln\frac{7}{4} + \frac{3\sqrt{3}}{8}}{\sqrt{3}/2} - \frac{-\ln\frac{7}{4} + \frac{3\sqrt{3}}{8}}{\sqrt{3}/2}}{\sqrt{3}} = \frac{\frac{2\ln\frac{7}{4}}{\sqrt{3}/2}}{\sqrt{3}} = \frac{4}{3}\ln\frac{7}{4}$$

$$P_2(x) = \left(\ln\frac{7}{4} - \frac{3\sqrt{3}}{8}\right) + \left(\frac{-\ln\frac{7}{4} + \frac{3\sqrt{3}}{8}}{\sqrt{3}/2}\right)\left(x + \frac{\sqrt{3}}{2}\right) + \frac{4}{3}\ln\frac{7}{4}\left(x + \frac{\sqrt{3}}{2}\right)x =$$

$$= \frac{2}{4}x + \frac{4}{3}\ln\frac{7}{4}x^2$$

В узлах:

$x_0 = -\frac{\sqrt{3}}{2}$: $P_2\left(-\frac{\sqrt{3}}{2}\right) = -\frac{3\sqrt{3}}{8} + \ln\frac{7}{4} = f\left(-\frac{\sqrt{3}}{2}\right)$

$x_1 = 0$: $P_2(0) = 0 = f(0)$

$x_2 = \frac{\sqrt{3}}{2}$: $P_2\left(\frac{\sqrt{3}}{2}\right) = \frac{3\sqrt{3}}{8} + \ln\frac{7}{4} = f\left(\frac{\sqrt{3}}{2}\right)$

В серединах:

$x = -\frac{\sqrt{3}}{4}$: $P_2\left(-\frac{\sqrt{3}}{4}\right) \approx -0{,}185$, $f\left(-\frac{\sqrt{3}}{4}\right) \approx 0{,}091$

$$\Rightarrow \left|P_2\left(-\frac{\sqrt{3}}{4}\right) - f\left(-\frac{\sqrt{3}}{4}\right)\right| \approx 0{,}276$$

$x = \frac{\sqrt{3}}{4}$: $P_2\left(\frac{\sqrt{3}}{4}\right) \approx 0{,}465$, $f\left(\frac{\sqrt{3}}{4}\right) \approx 0{,}253$

$$\Rightarrow \left|P_2\left(\frac{\sqrt{3}}{4}\right) - f\left(\frac{\sqrt{3}}{4}\right)\right| \approx 0{,}212$$